Optoelectronic lab

Experiment 4: Light-emitting diodes - basics, radiation and optical outcoupling

1 Introduction

In this practical experiment you will learn about important characteristics of light-emitting diodes.

Since a light-emitting diode is an electrically powered emitter of light (photons) in the broadest sense, it is characterized by the following data:

- Emission spectrum (color), i.e. the intensity as a function of wavelength. LEDs are now commercially available from the UV (340 nm) to the mid-IR range (5000 nm).
- Quantum efficiency, which is characterized by the number of emitted photons in relation to the number of injected electrons.

2 Basics

2.1 Electrical properties

The electronic properties of a pn contact are of central importance for light-emitting diodes. These are only briefly described here; for further information, please refer to the literature.

If an n-doped semiconductor is brought into contact with a p-doped semiconductor, the Fermi levels are equalized in thermodynamic equilibrium. This results in the band diagram shown in Fig. 1.

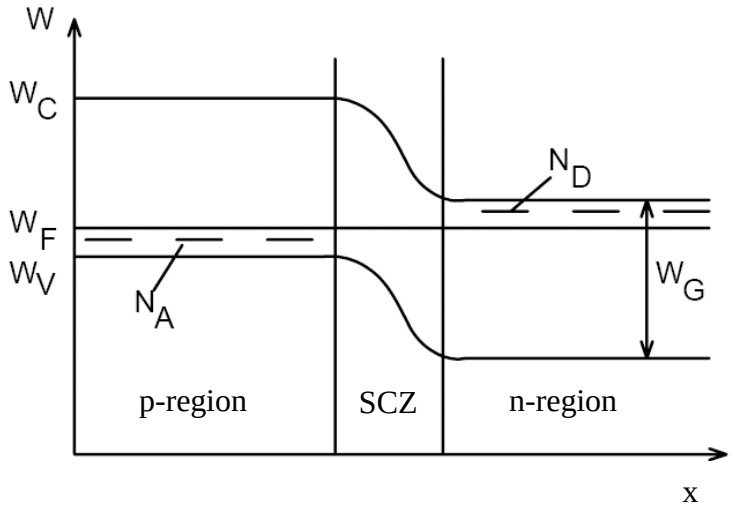


Fig. 1: *Band diagram at the pn junction*

Between the n-doped orbital region with electrons as majority carriers and the p-doped orbital region with holes as majority carriers, a space charge zone is formed that is depleted of free charge carriers. Fig. 2 shows the current-voltage (I-V) characteristic of a pn junction, which is described by the equation

$$I = I_s\left(e^{U/U_T} - 1\right) \tag{1}$$

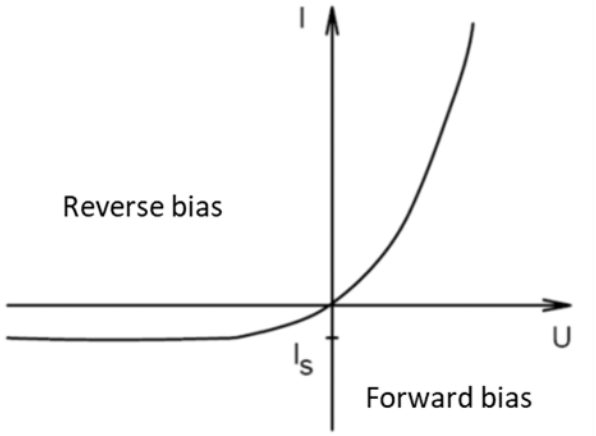


***Fig. 2**: I-V characteristic of a p-n junction*

Here is: $U_T = k \cdot T/e$ and $I_S = eA\left(\frac{n_i^2 L_p}{N_D \tau_p} + \frac{n_i^2 L_n}{N_A \tau_n}\right)$,

with
- A: Cross section diameter of the diode
- I_S: Saturation current
- n_i: Intrinsic concentration
- N_D, N_A: Doping concentrations
- τ_p, τ_n: Charge carrier lifetimes
- $L_p = \sqrt{D_p \tau_p}$
- $L_n = \sqrt{D_n \tau_n}$
- D_p, D_n: Diffusion coefficients
- e: Elementary charge

2.2 Structure

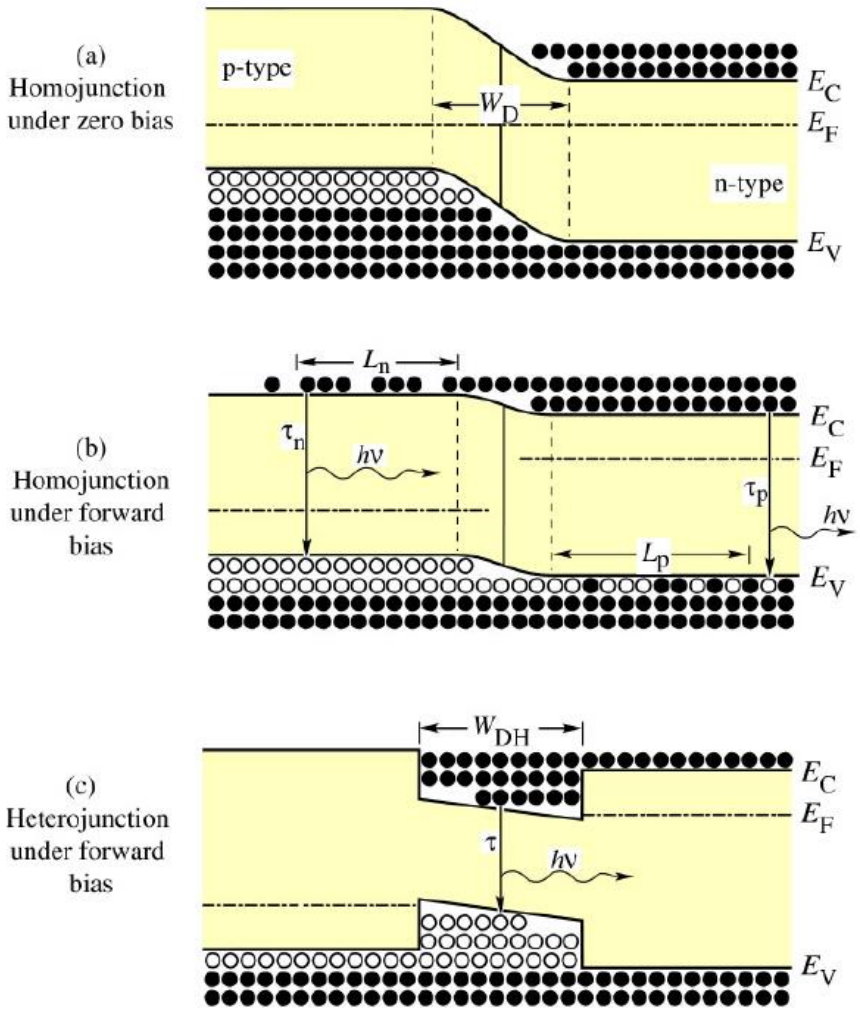


Fig. 3: *Comparison of homo- and heterostructures. From [1]*

Virtually all LEDs are based on a double heterostructure, i.e. material with a small band gap embedded in two barriers (material with a high band gap).

The advantage is immediately apparent from Fig. 3: The region in which the charge carriers recombine is restricted from the width W_D (corresponding to the diffusion length, which is > 10 µm for GaAs, for example) to an active region of width W_{DH} (in the sub-µm range, in the case of quantum films even in the nm range), so that recombination becomes much more effective.

There is a second effect: the light generated in the active region is not re-absorbed by the substrate or cover layer, as these are transparent to the light due to their larger band gap.

2.3 Emission spectrum

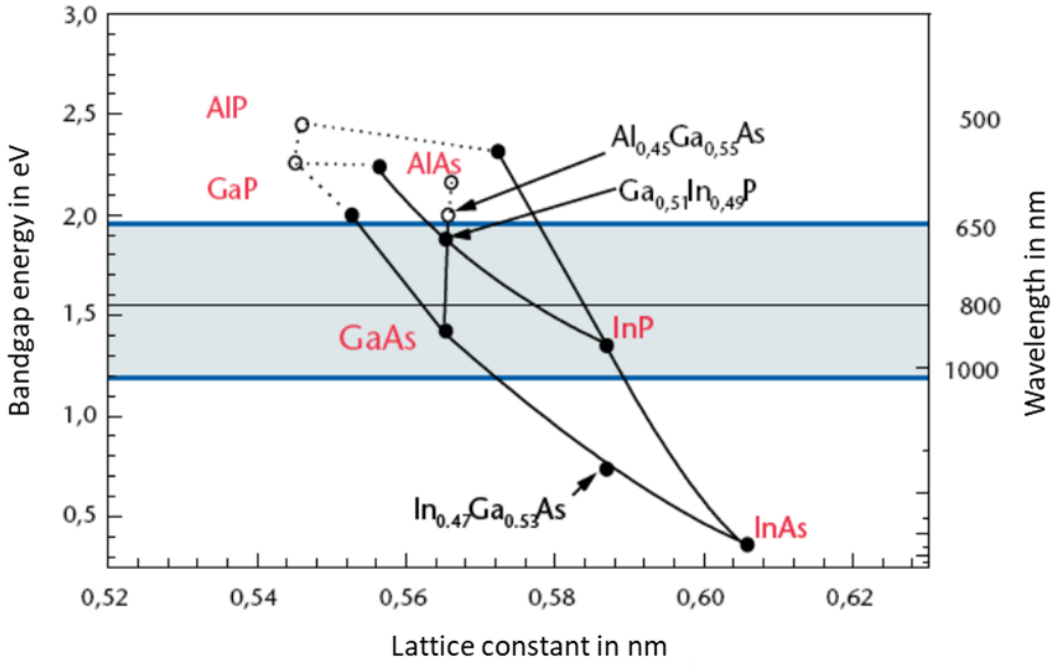


***Fig. 4**: Material systems and emission energies for various GaAs and GaP-based LEDs. The emission energies are given in eV, the unit commonly used in spectroscopy, and can be converted into wavelengths according to:* $\lambda[nm] = \frac{1240}{E[eV]}$ *. Source: Ferdinand-Braun-Institut für Höchstfrequenztechnik, Berlin*

The light from the LED is generated by the recombination of electron-hole pairs in the active zone.

As LEDs can be based on a variety of material systems, they can emit light in many different colors. LEDs in the yellow-red and infrared range (based on GaP and GaAs compounds, see Fig. 4) are particularly well researched. In the last 15 years, blue LEDs based on GaN have gained enormously in importance.

The theoretical emission spectrum follows the following law

$$I(E) \propto \sqrt{E - E_g}\, \exp(-E / kT) \qquad (2)$$

The band gap E_g is a decisive factor. Quantum structures (quantum films) are also frequently used, in which case the quantization energy must also be considered. The root term describes the density of states for electron-hole pairs in the bulk material, whose recombination releases the energy E. The exponential term originates from the (thermal) Boltzmann distribution. (For details, please refer to the literature). The resulting emission spectrum is sketched in Fig. 5. Important is the line width, which for room temperature is theoretically a few 10 meV.

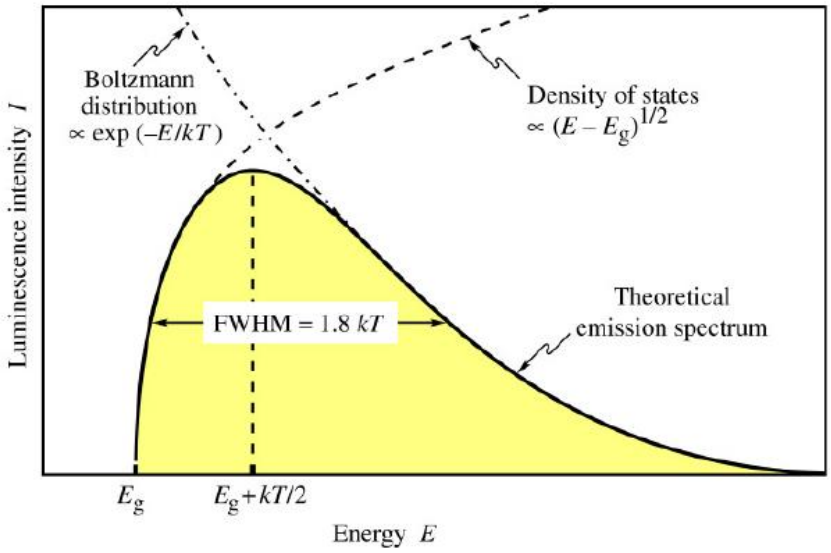


***Fig. 5**: Theoretical emission spectrum of an LED. From [1]*

2.4 Quantum efficiency of LEDs

Two factors impair the quantum efficiency (i.e. the conversion of injected electrical power into radiated power in the form of photons): Firstly, not all injected electrons/hole pairs are converted into photons; in non-radiative processes, heat is generated instead. Secondly, not all the photons generated leave the LED to the outside.

The following variables characterize the quantum efficiency:

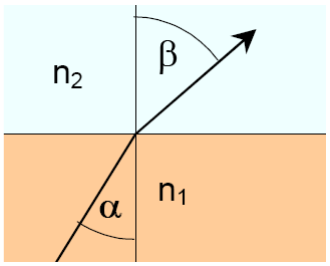
- Internal quantum efficiency: $\eta_{\text{int}} = \frac{number\ of\ generated\ photons\ in\ active\ region/s}{number\ of\ injected\ electrons/s}$
- External quantum efficiency: $\eta_{\text{ext}} = \frac{number\ of\ emitted\ photons\ into\ free\ space\ /s}{number\ of\ injected\ electrons/s}$
- The output coupling efficiency $\eta_{eff} = \frac{\eta_{ext}}{\eta_{\text{int}}}$ describes the proportion of photons generated that are actually emitted into free space and is therefore a key parameter of the LED.

The internal quantum efficiency can be improved, for example, by using the double heterostructures described above, which increase the probability of recombination. In addition, quantum films are often used as active regions; here the efficiency is further increased.

However, since the main problem for poor efficiency is a low outcoupling efficiency, this topic will be discussed in more detail in the following chapter.

3 Improving external quantum efficiency

3.1 Reflection of light at interfaces



When light is refracted at a boundary surface, the angle of incidence α and the angle of reflection β are known to be related to the refractive indices via Snellius' law of refraction:

$$\frac{n_1}{n_2} = \frac{\sin\beta}{\sin\alpha} \qquad (3)$$

Fresnel's formulae describe the relationship between the intensities of the incident, reflected and transmitted (refracted) light at a boundary surface.

- The reflection coefficient $R = \frac{reflected\ power}{incident\ power}$ is defined, and the polarization of the light must also be taken into account.
  - R_S describes the reflection of light that is polarized perpendicular to the drawing plane,
  - R_P the reflection of light that is polarized parallel to the drawing plane.

In the case considered here (non-magnetic materials, negligible absorption within the component), the following applies

$$R_S = \frac{\sin^2(\alpha-\beta)}{\sin^2(\alpha+\beta)},\ R_P = \frac{\tan^2(\alpha-\beta)}{\tan^2(\alpha+\beta)} \qquad (4)$$

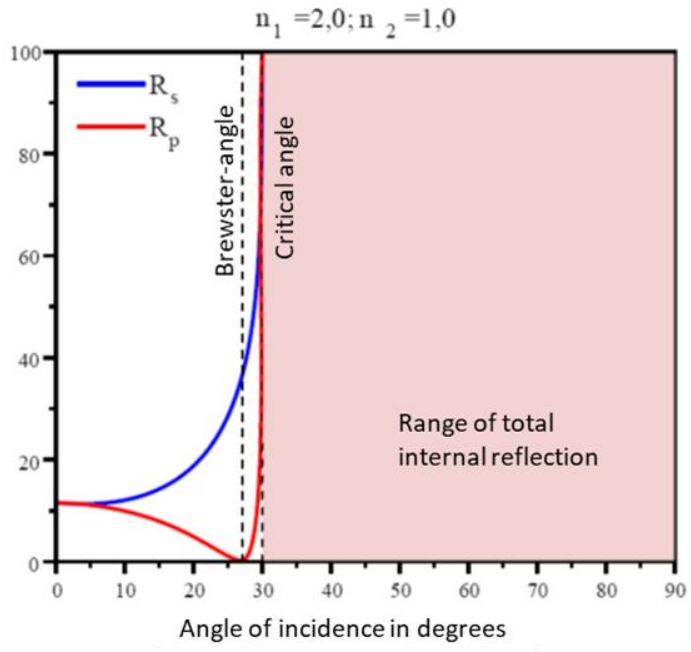


***Fig. 6**: Reflection coefficients as a function of the angle of incidence (Fresnel reflection)*

The following applies to unpolarized light

$$R = \frac{R_S + R_P}{2} \qquad (5)$$

However, equation (4) is not defined for the case of vertical irradiation (i.e. α=β=0).

In this case, $R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$. (6)

- Derive this equation from the law of refraction and (4) by using the relationship sin(x)≈ tan(x)≈ x for small angles.

This means that a considerable proportion of the light generated is reflected back into the semiconductor.

The total reflection for the angles $\alpha > \phi_C$ is more decisive for the overall low degree of outcoupling, whereby the critical angle ϕ_C applies:

$$\sin(\phi_C) = \frac{n_2}{n_1} \text{ or, assuming that } \phi_C \text{ is small, simply } \phi_C \approx \frac{n_2}{n_1}$$

This means that only a cone with an opening angle of ϕ_C contributes to the emission.

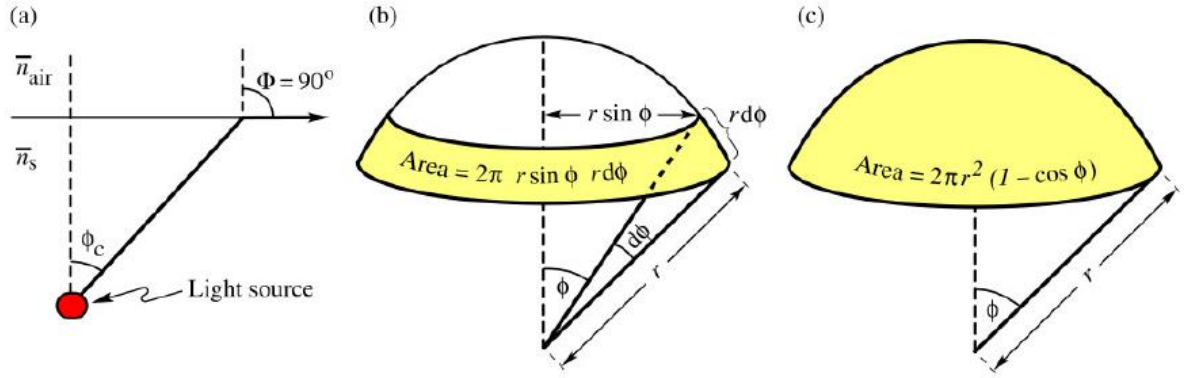


Fig. 5.3. (a) Definition of the escape cone by the critical angle ϕ_c. (b) Area element d*A*. (c) Area of calotte defined by radius *r* and angle ϕ_c.

Fig. 7: Emergence cone of the generated light in the LED. From [1]

In order to quantify this contribution, the ratio of the solid angle of the exit cone to the total solid angle must be calculated. The solid angle results from Fig. 7 as

$$A = \int dA = \int_0^{\phi_C} 2\pi r \sin\phi r d\phi = 2\pi r^2 (1 - \cos\phi_C)$$

This area must be related to the spherical surface $4\pi r^2$; for the proportion of outcoupled radiation, it must also be multiplied by the Fresnel reflection losses (*1-R*) and the result is

$$\eta_{eff} = \frac{2\pi r^2 (1 - \cos\phi_C)}{4\pi r^2} \cdot (1 - R) = \frac{1 - R}{2} (1 - \cos\phi_C) \quad (7)$$

This formula is also not entirely accurate! The fact that the Fresnel reflection coefficient R is angle-dependent has been neglected.

- The approximation is still quite good, why? (Tip: Take a look at Fig. 6)
- GaAs has a refractive index of n_1 =3.5. What is the outcoupling efficiency at the interface to air (n_2 =1)?

With $\cos(\phi_C) \approx 1 - \frac{\phi_C^{\,2}}{2}$ and $\phi_C \approx \frac{n_2}{n_1}$, (7) is even easier to write as

$$\eta_{eff} \approx \frac{1-R}{4}\left(\frac{n_2}{n_1}\right)^2 \tag{8}$$

3.2 Lambert spotlight

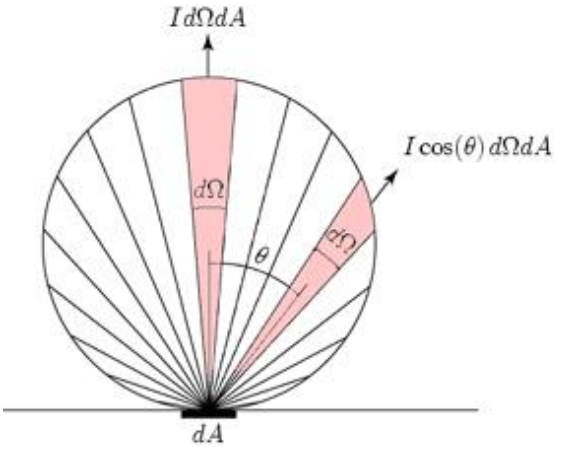


***Fig. 8**: Lambert radiator*

A diffuse, planar radiator (such as a frosted screen in front of a lamp) does not emit isotropically (i.e. does not provide the same radiant flux (energy/s) in every direction), but the radiant flux depends on the radiation angle Θ (in relation to the surface normal):

$$I(\Theta) \propto \cos(\Theta)$$

This results directly from the perspective foreshortening of a surface viewed "obliquely", i.e. from the projection of the surface normal onto the direction of observation. Such a radiator is called a *Lambertian radiator*.

After discussing the angular range in the LED from which the emitted radiation originates, we will briefly examine how the angular dependence of the *emitted* intensity behaves. We ignore Fresnel reflection in this section.

For our considerations, we assume that (as described above) the angle of incidence α is small, so that sin α ≈ α. For the angle of reflection β this does not apply, so Snellius' law can be written as

$$n_{HL}\alpha = n_{Luft}\sin\beta \tag{9}$$

where n_{HL}, n_{Luft} are the corresponding refractive indices. Light from a differential angle $\delta\alpha$ (in the semiconductor) is correspondingly refracted into an angle dβ (in air), and by differentiating (9) we obtain

$$d\beta = \frac{n_{HL}}{n_{Luft}}\frac{1}{\cos\beta}d\alpha \tag{10}$$

A surface element in the area of the outgoing radiation, i.e. in air, is (using the last two equations, see also Fig. 7 again – here ϕ is now replaced by β)

$$dA_{Luft} = 2\pi r\sin\beta\, r\, d\beta = 2\pi r^2\sin\beta\,\frac{n_{HL}}{n_{Luft}}\frac{1}{\cos\beta}d\alpha = 2\pi r^2\,\frac{n_{HL}^{\,2}}{n_{Luft}^{\,2}}\frac{1}{\cos\beta}\alpha\, d\alpha \tag{11}$$

A surface element in the semiconductor is defined as (where again sin α ≈ α is used)

$$dA_{HL} = 2\pi r\sin\alpha\, r\, d\alpha \approx 2\pi r^2\alpha\, d\alpha \tag{12}$$

It must now be taken into account that the intensity I_S (measured in W/m^2) inside the semiconductor on the surface dA_{HL} is equal to the intensity I_{Luft} outside the semiconductor on the surface dA_{Luft} , i.e.

$$I_{HL} dA_{HL} = I_{Luft} dA_{Luft} . \tag{13}$$

If we also take into account that the power P generated within the semiconductor at a distance r from the source produces the intensity $I_S = P/(4\pi r^2)$, we finally obtain the intensity I_{Luft} as a function of β:

$$I_{Luft} = \frac{P}{4\pi r^2} \frac{n_{Luft}^{\ 2}}{n_{HL}^{\ 2}} \cos\beta \tag{14}$$

It is clear from this that a flat LED has the emission of a Lambertian spotlight.

3.3 Improving light extraction

The outcoupling efficiency can be improved if the semiconductor-air interface is replaced by a semiconductor-epoxy interface followed by an epoxy-air interface. This has several advantages: Firstly, the smaller refractive index jump between semiconductor and epoxy increases the critical angle ϕ_C. Secondly, epoxy can be formed much more easily than a semiconductor, so that the epoxy-air interface is largely free of total internal reflection. Fig. (9) shows the increase in outcoupling efficiency with the introduction of a so-called "epoxy dome" as a function of the refractive index of the epoxy for two different semiconductors. However, the efficiency can only be approximately tripled. The modified geometry results in a deviation from the Lambertian radiation pattern.

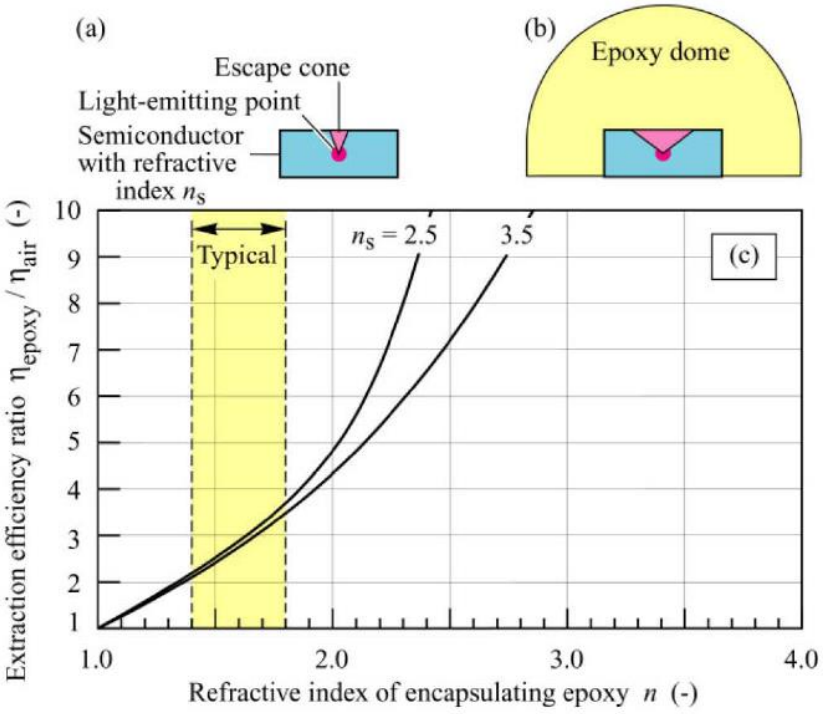


Fig. 9: *Increasing outcoupling efficiency with epoxy. From [1]*

- Consider qualitatively (drawing) how to shape the epoxy dome so that you emit at the smallest possible angle.

Geometries for highly efficient light extraction are currently under intensive investigation.

One example is the so-called "truncated inverted pyramid" (TIP) geometry, which was optimized using numerical methods (ray tracing). The current distributions are also included in these calculations; for example, the thick n-GaP layer enables a widening of the current flow and therefore less shadowing through the contact.

Other concepts use textured ("intentionally rough") surfaces through which the light can be coupled out. Mirror coatings prevent photons from being lost through the back.

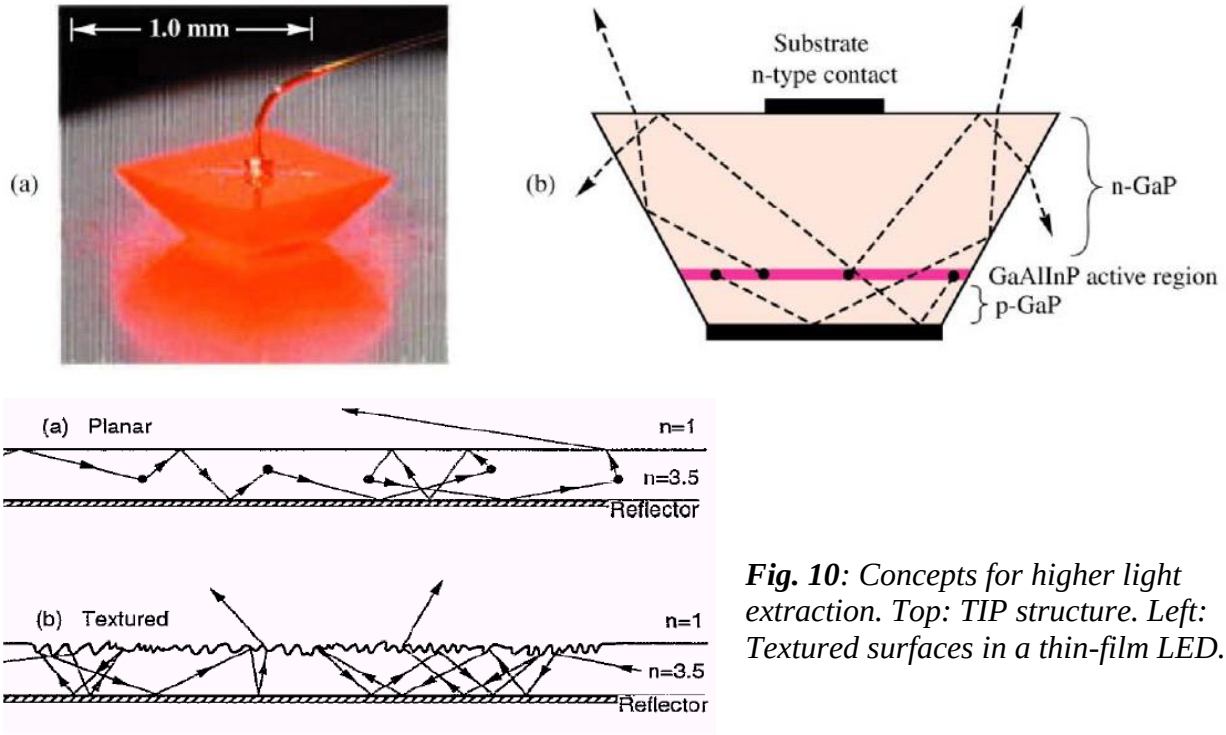


***Fig. 10**: Concepts for higher light extraction. Top: TIP structure. Left: Textured surfaces in a thin-film LED.*

4 Diodes as lighting elements

4.1 Luminous flux

For LEDs used as signal lamps or for lighting, the emitted power (photons/s) is a clearly defined and measurable quantity, but it does not take into account the wavelength dependence of the sensitivity of the human eye.

The *luminous flux* is defined as

$$\Phi_{vis} = K_m \int_{380nm}^{780nm} V(\lambda) \frac{\partial \Phi_e}{\partial \lambda} d\lambda$$ where Φ_e stands for the radiation flux (energy/s) in watts, Φ_{vis}

for the luminous flux in lumens, $V(\lambda)$ for the brightness sensitivity curve of the human eye and K_m = 683 lm/W is a weighting factor; this results from the maximum luminous efficacy (Φ_{vis}/Φ_e) at λ=555nm.

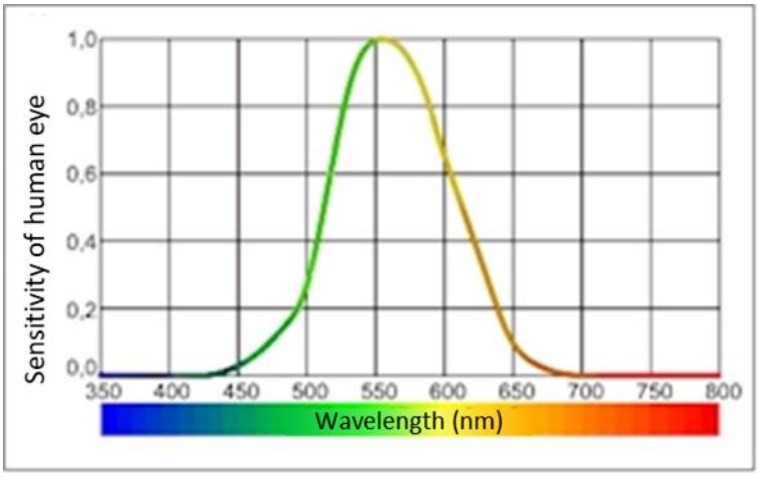


***Fig. 11:** Brightness sensitivity of the human eye.*

4.2 Radiation characteristics

The radiation characteristic of an LED indicates the luminous intensity as a function of the direction of radiation. It depends on the material and the shape of the epoxy dome and is described by a so-called luminous intensity distribution curve (LID), which is usually specified in polar coordinates. The half beam angle of an LED, as defined, is the angle at which the intensity is reduced to half the maximum intensity. An example is shown in the following image.

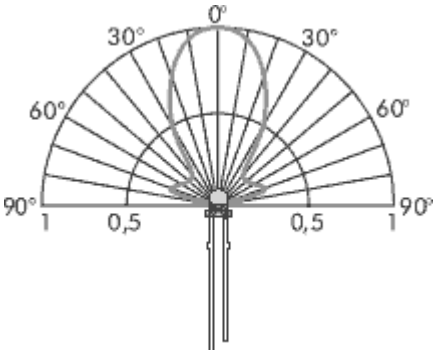


Fig. 12: *Characteristics of an LED with a beam angle of 60°.*

4.3 White LEDs

White light can be generated by the additive color mixing of different colored LEDs. In the simplest case, a red, a blue and a green LED are combined, or even just two LEDs with complementary colors (such as blue and yellow). This light looks white, but in contrast to the continuous solar spectrum, broad frequency ranges are missing. This often leads to poor color rendering of the illuminated objects.

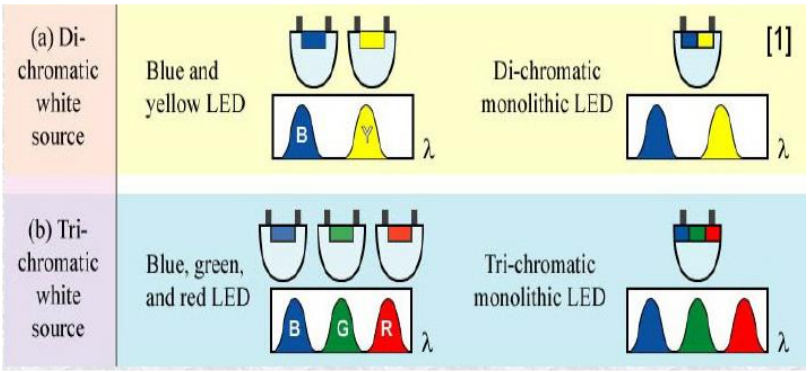


Fig. 13: *Principle of 2 or 3-color white light LEDs*

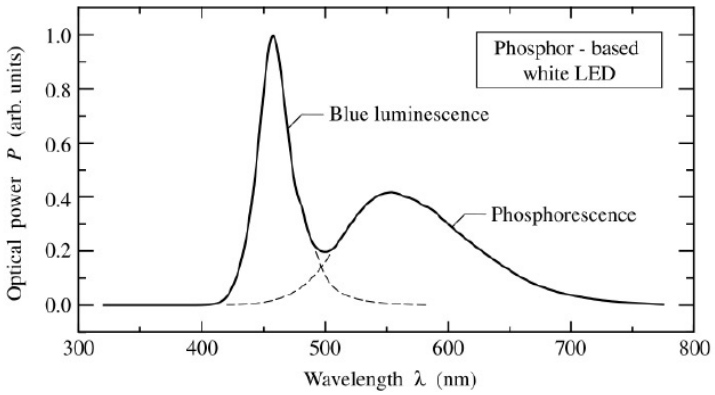


Fig. 14: *Principle of white light LEDs using a phosphor converter.*

Another approach uses LEDs that combine blue electroluminescence with a phosphor (often Ce ions embedded in yttrium aluminum garnet (YAG)) that emits broadband light in the green and red ranges. Intensive research is also being carried out in this area in order to produce both "natural-looking" and bright white light. However, unlike tricolor LEDs, it is not possible to match the colors.

5 Tasks

In the practical experiment, you will be given a series of diodes with different colors, output powers and radiation characteristics.

For measurements you can use:

- A current/voltage source to supply the diodes
- A multimeter for measuring current and voltage
- A spectrometer with an optical fiber, connected to the measuring computer, which enables you to record the spectra and intensities.
- An integrating sphere; this is a hollow sphere whose inner coating consists of materials with the best possible diffuse reflectance. Barium sulphate ($BaSO_4$) is often used. The diameter of the inlet and outlet openings is significantly smaller than the inner diameter of the sphere, so that only light that has previously been reflected many times on the inner surface reaches the outlet plane. The exit aperture therefore fulfills Lambert's law almost perfectly. As the openings are small, hardly any light is lost and absolute intensities can be compared, irrespective of the radiation characteristics of the LED.

Please carry out the following measurements for various diodes (selected by the lab supervisor) and discuss the results (do not exceed a current of 30 mA in your measurements, as this can destroy the diodes):

- Measure the current/voltage characteristics of several LEDs and compare them. Record enough measuring points to be able to carry out a reasonable evaluation. What causes the differences in the I-V curves of the various LEDs?
- Record the emission spectra of the LEDs. Determine the emission maxima and half-width from the recorded spectra. How are the characteristic curves related to the emitted color?
- Measure the radiant flux (the "light output") as a function of the current using the integrating sphere and plot the measured values in P-I curves. Compare "super-bright" LEDs with "normal" LEDs. How does the coupled electrical power compare to the radiant flux?
- Mount the diodes on a rotatable holder. Consider how the diodes need to be mounted on the bracket in order to measure only the angular dependence and not have to consider any additional displacement of the diodes. Record the radiation characteristics (radiation flux as a function of angle) for different diodes and compare them.
- Compare the ("subjective") color rendering of different white LEDs. To do this, display the respective spectra and discuss the differences in the resulting spectra in relation to the color rendering you perceived with your eyes.

When discussing the results in the test protocol, the following should be addressed in particular in each of the points to be measured:

- What results do I expect according to theory and what results do I get in reality?
- How can any deviations between theory and reality be explained (error analysis)?

6 Literature:

- [1] E.F. Schubert "Light Emitting Diodes", many of the illustrations used here were taken from this book. See also http://www.lightemittingdiodes.org
- [2] M. Fukuda: Optical Semiconductor Devices